



Stellenbosch

UNIVERSITY
IYUNIVESITHI
UNIVERSITEIT

OPERATIONS RESEARCH 415 PART IV: QUEUING THEORY

LECTURE 33: THE QUEUING SYSTEM $M/G/s/GD/s/\infty$

Presenter: Dr Philip Venter

Department of Industrial Engineering

Blocked Customers Cleared (pp. 1112–1113)

In many queuing systems, an arrival who finds all servers occupied is, for all practical purposes, lost to the system.

For example, a person who calls an airline for a reservation and gets a busy signal will probably call another airline.

If arrivals who find all servers occupied leave the system, we call the system a *blocked customers cleared (BCC) system*.

Assuming that inter-arrival times are exponentially distributed, such a system may be modeled as an $M/G/s/GD/s/\infty$ system for which L , W , L_q , and W_q are of limited interest.

For example, since a queue can never occur, $L_q = W_q = 0$.

Furthermore, if we let $1/\mu$ be the mean service time and λ the arrival rate, then $W = W_s = \frac{1}{\mu}$.

Blocked Customers Cleared (pp. 1112–1113)

In most BCC systems, primary interest is focused on the fraction of all arrivals who are turned away.

Since arrivals are turned away only when s customers are present, a fraction π_s of all arrivals will be turned away.

Hence, an average of $\lambda\pi_s$ arrivals per unit time will be lost to the system.

Since an average of $\lambda(1 - \pi_s)$ arrivals per unit time will actually enter the system, we may conclude that

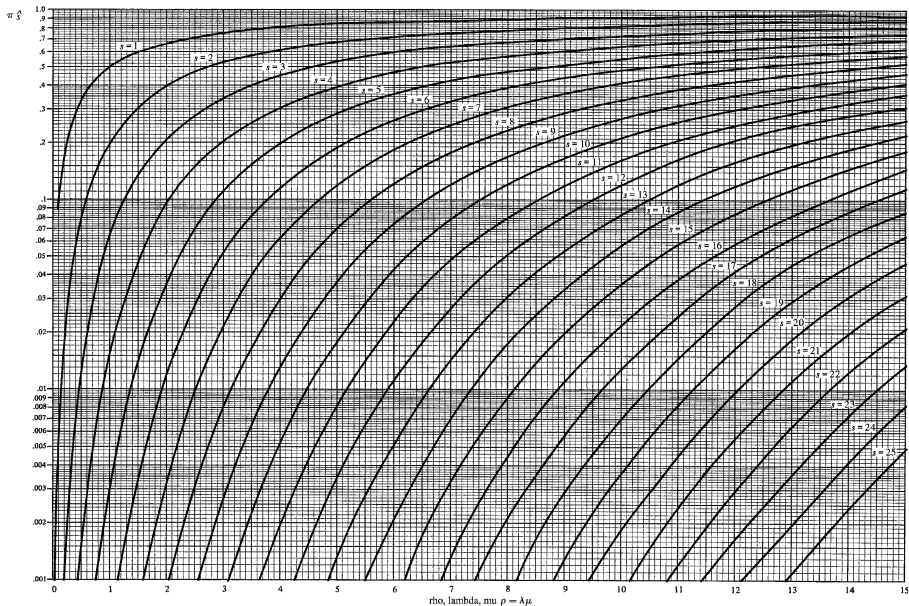
$$L = L_s = \frac{\lambda(1 - \pi_s)}{\mu}.$$

For an $M/G/s/GD/s/\infty$ system, it can be shown that π_s depends on the service time distribution only through its mean ($1/\mu$).

In other words, any $M/G/s/GD/s/\infty$ system with an arrival rate λ and mean service time of $1/\mu$ will have the same value of π_s .

How is this value of π_s determined?

Blocked Customers Cleared (pp. 1112–1113)



Ambulance Calls (Example 15, p. 1112)

An average of 20 ambulance calls per hour are received by Gotham City Hospital.

An ambulance requires an average of 20 minutes to pick up a patient and take the patient to the hospital.

The ambulance is then available to pick up another patient.

How many ambulances should the hospital have to ensure that there is at most a 1% probability of not being able to respond immediately to an ambulance call?

(Assume that inter-arrival times are exponentially distributed.)

The *I Ching* (Problem 7, p.1115)

On average, 26 patrons per year go to the I.U. library to borrow the *I Ching* (assume that interarrival times are exponential).

Borrowers who find the book unavailable leave and never return.

A borrower keeps a copy of the *I Ching* for an average of 4 weeks.

- 1 If the library has only one copy, what is the expected number of borrowers who will come to borrow the *I Ching* each year and find that the book is not available?
- 2 Suppose that each person who comes to borrow the *I Ching* and is unable to is considered to cost the library \$1 in goodwill. A copy of the *I Ching* lasts two years and costs \$11. A thief has just stolen the library's only copy. To minimize the sum of purchasing and goodwill costs over the next two years, how many copies of the *I Ching* should be purchased?